EC 571: Digital VLSI Circuit Design

Timing

Ajay Joshi

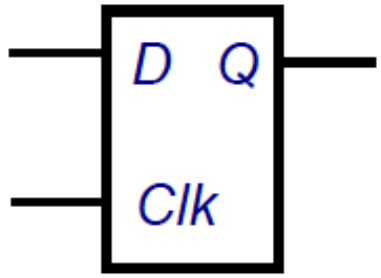
Acknowledgement - Some slides adapted from:
- J. Rabaey et al. Digital Integrated Circuits 2nd Ed.
- UC Berkeley EE141, UC Berkeley EE241, UC Berkeley EE251
- Stanford EE271

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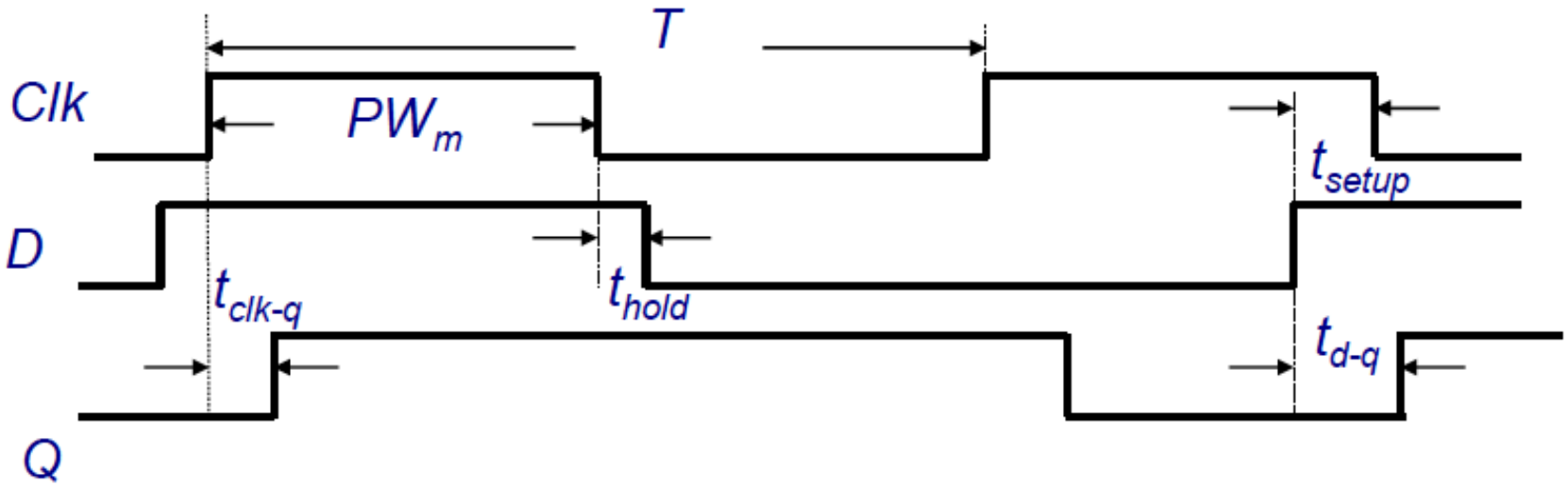
Last lecture

- Last lecture
  - Flip-flop and Latch Circuit Design
- Today's lecture
  - Timing
- Reading
  - Chapter 7

Latch parameters

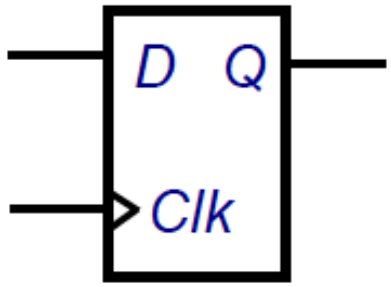


Delays can be different for rising and falling data transitions

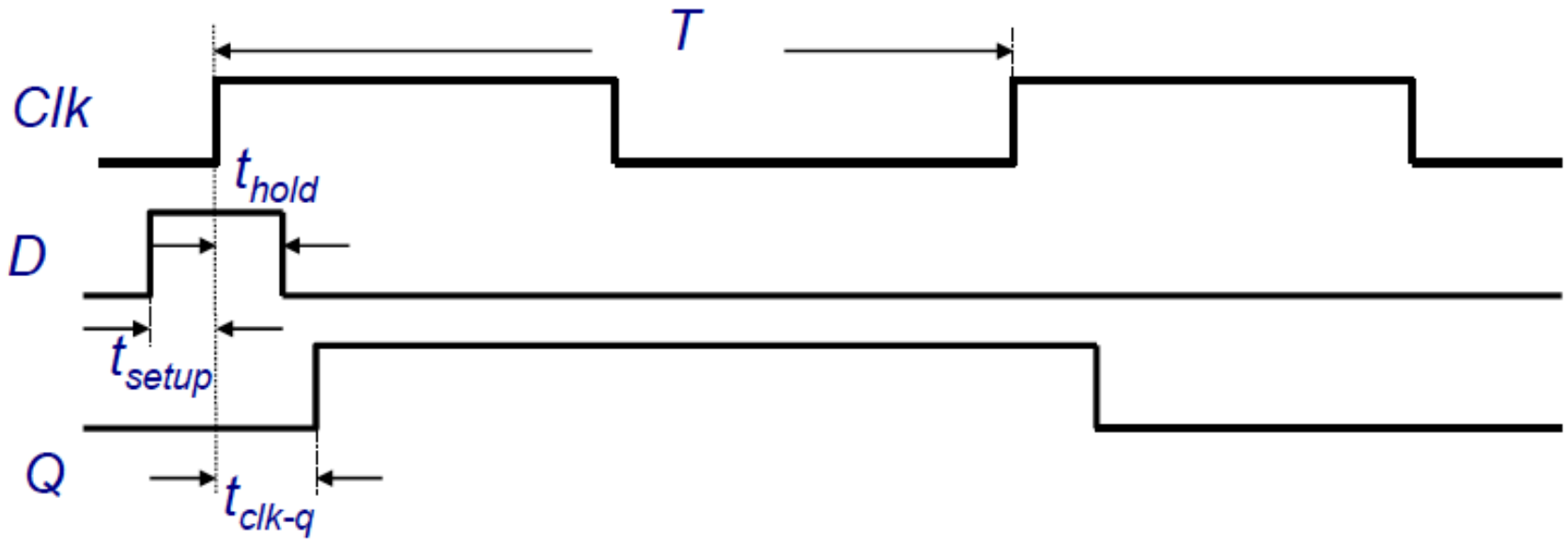


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Register parameters

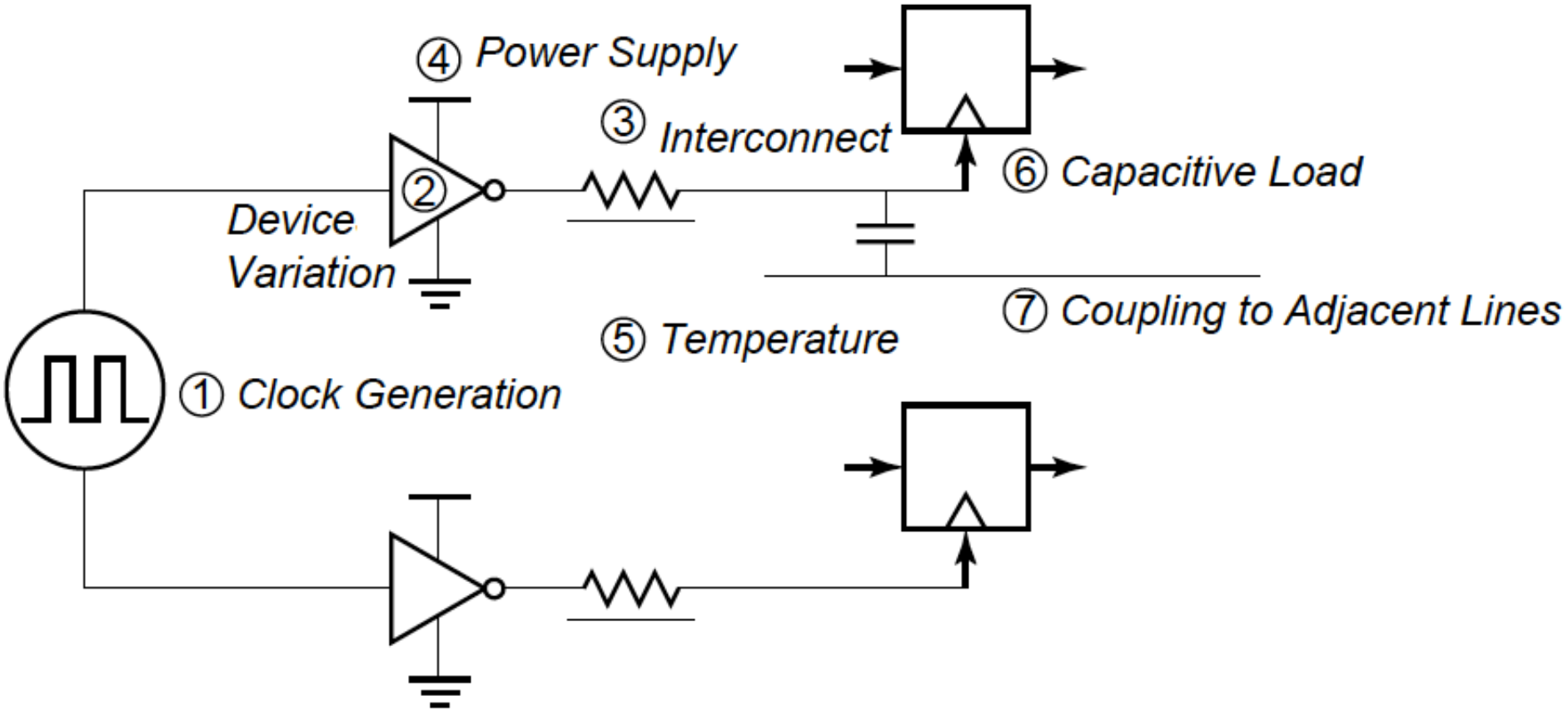


Delays can be different for rising and falling data transitions



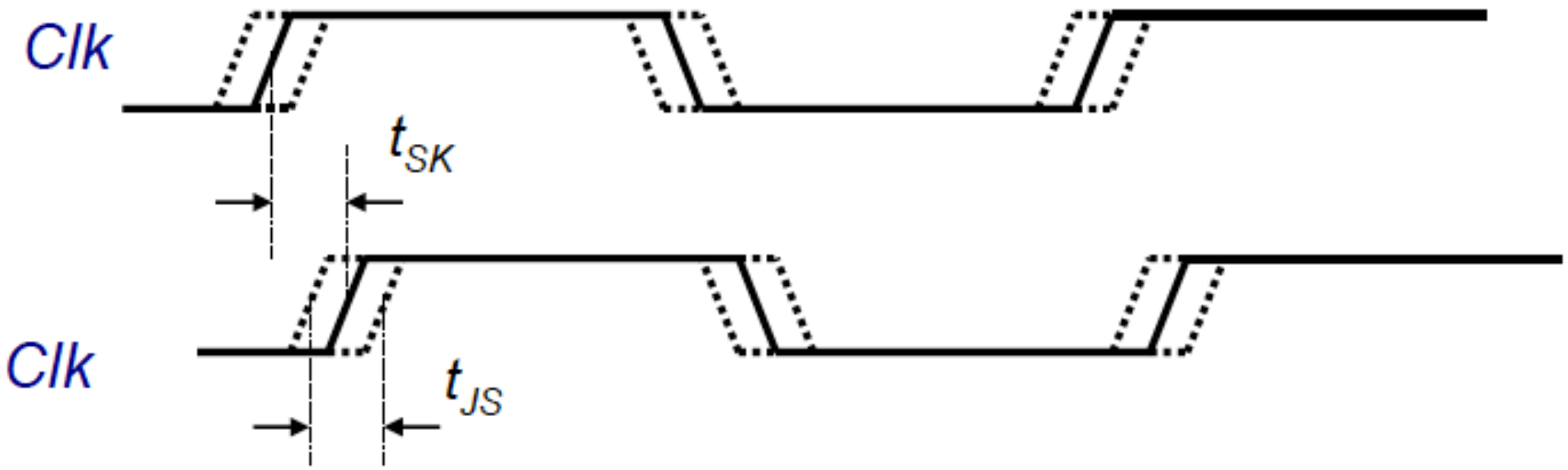
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Clock uncertainties



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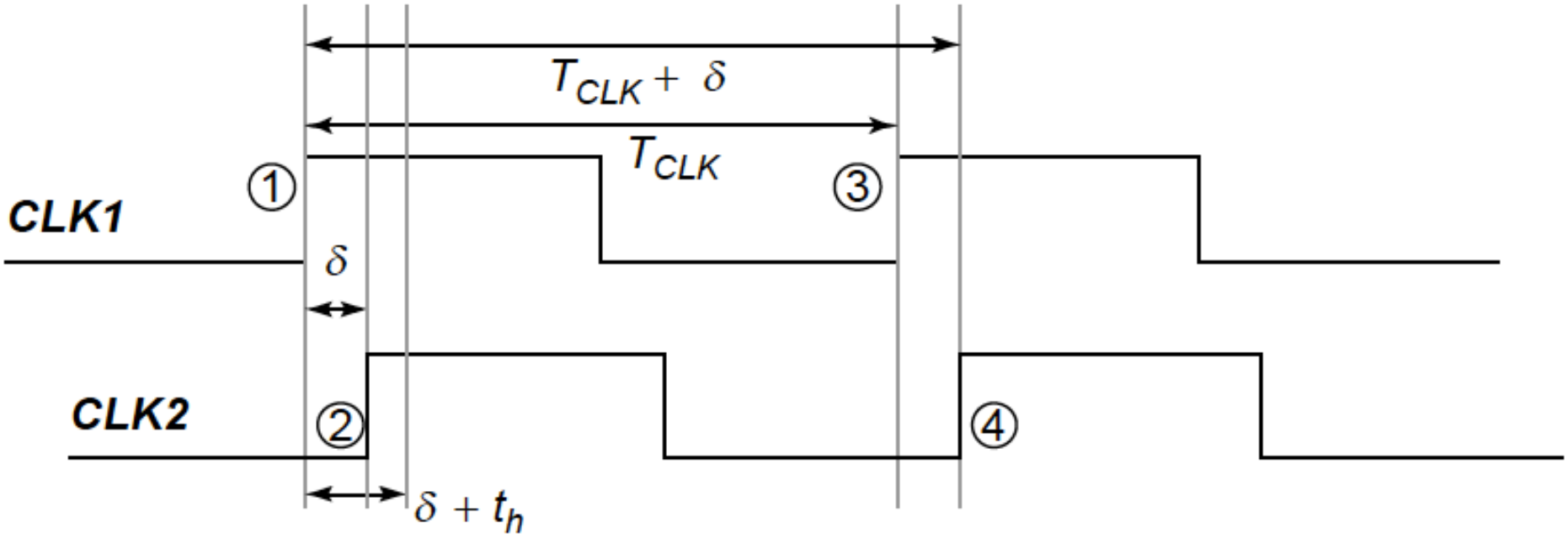
Clock skew and jitter



- Both skew and jitter affect the effective cycle time
- Only skew affects the race margin (usually)

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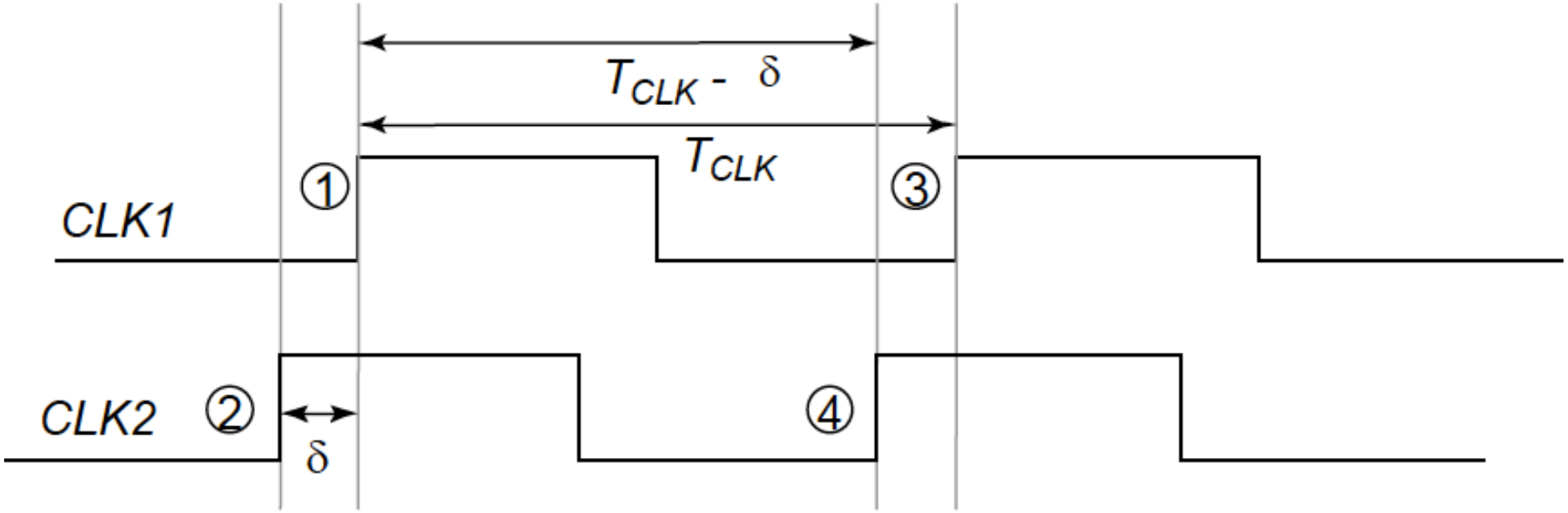
Positive skew



Launching edge arrives before receiving edge

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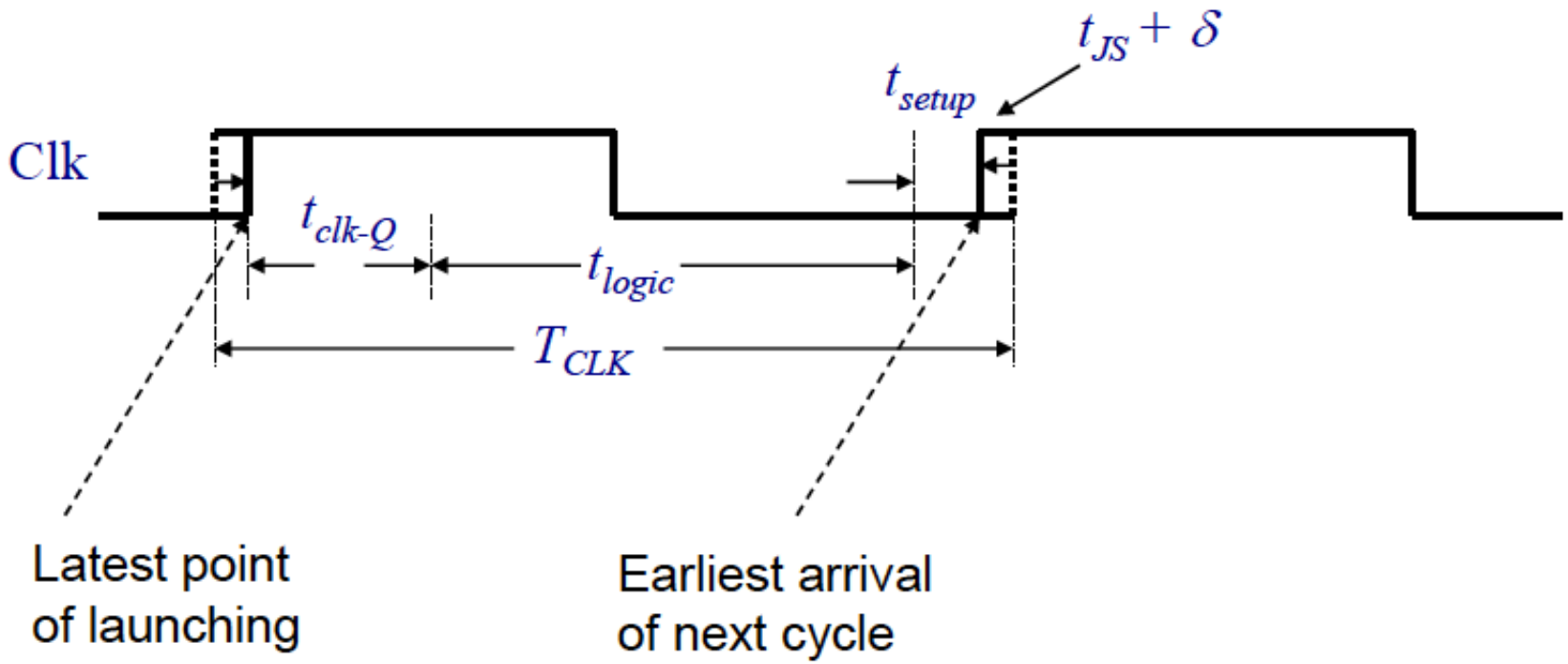
Negative skew



Launching edge arrives after receiving edge

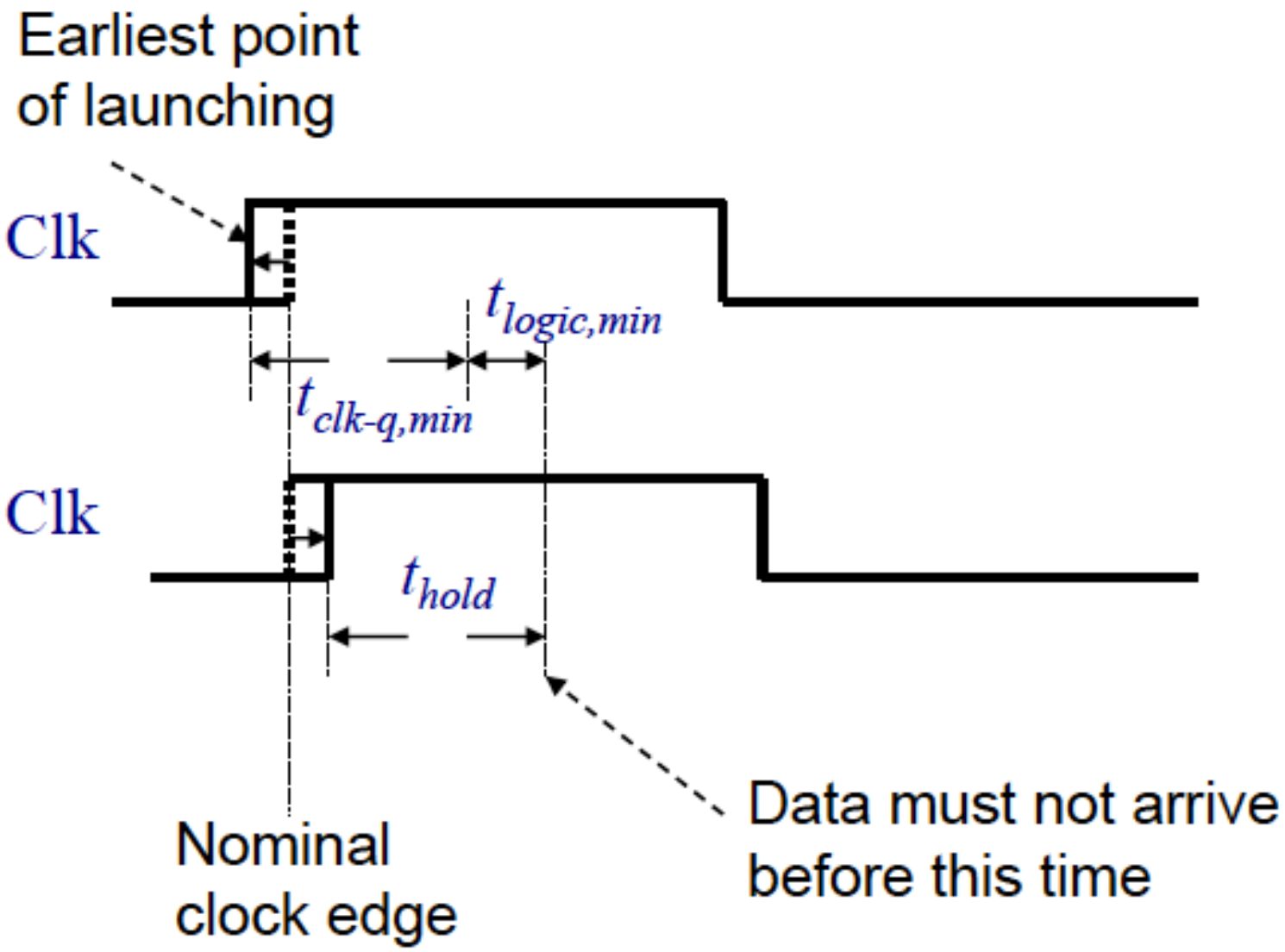
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Longest logic path in edge-triggered system



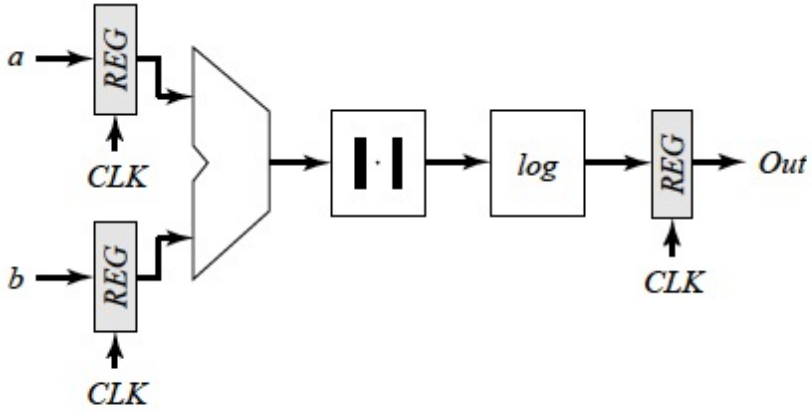
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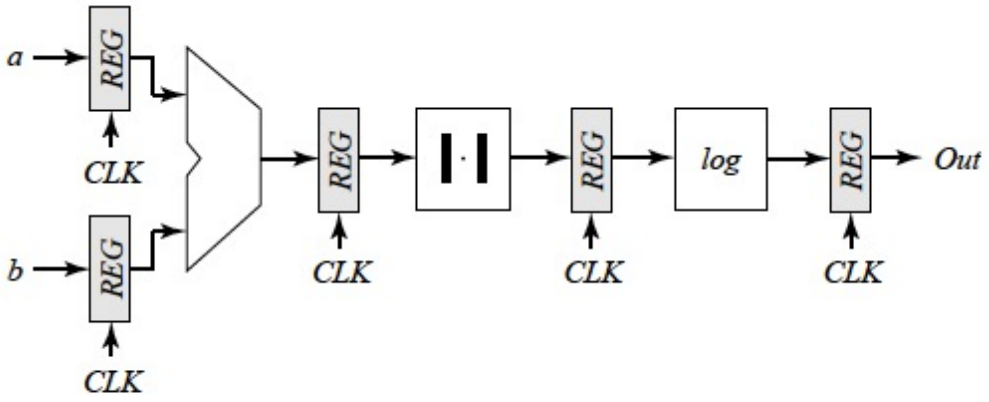
Shortest path



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Pipelining

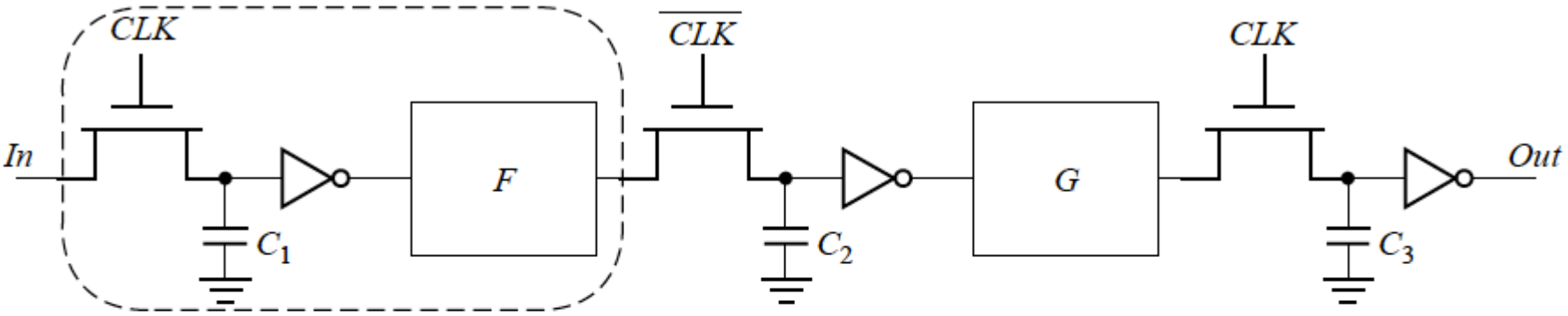


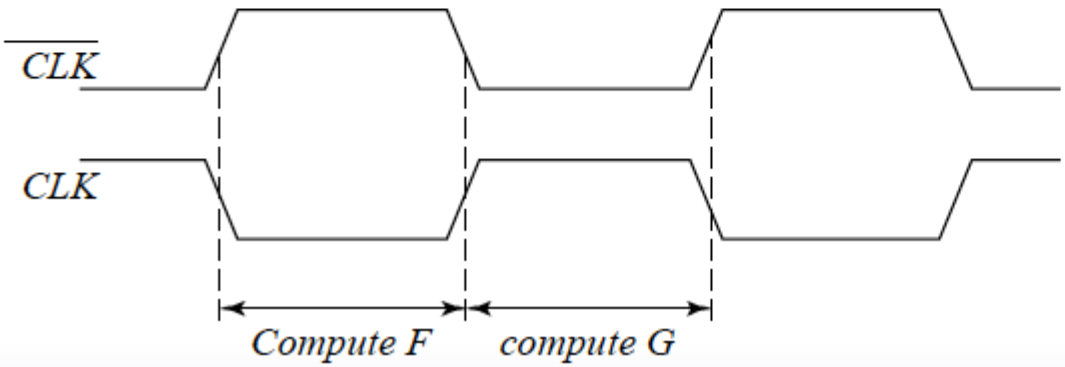


Clock Period	Adder	Absolute Value	Logarithm
1	$a_1 + b_1$		
2	$a_2 + b_2$	$\|a_1 + b_1\|$	
3	$a_3 + b_3$	$\|a_2 + b_2\|$	$\log(\|a_1 + b_1\|)$
4	$a_4 + b_4$	$\|a_3 + b_3\|$	$\log(\|a_2 + b_2\|)$
5	$a_5 + b_5$	$\|a_4 + b_4\|$	$\log(\|a_3 + b_3\|)$

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Latch-based clocking

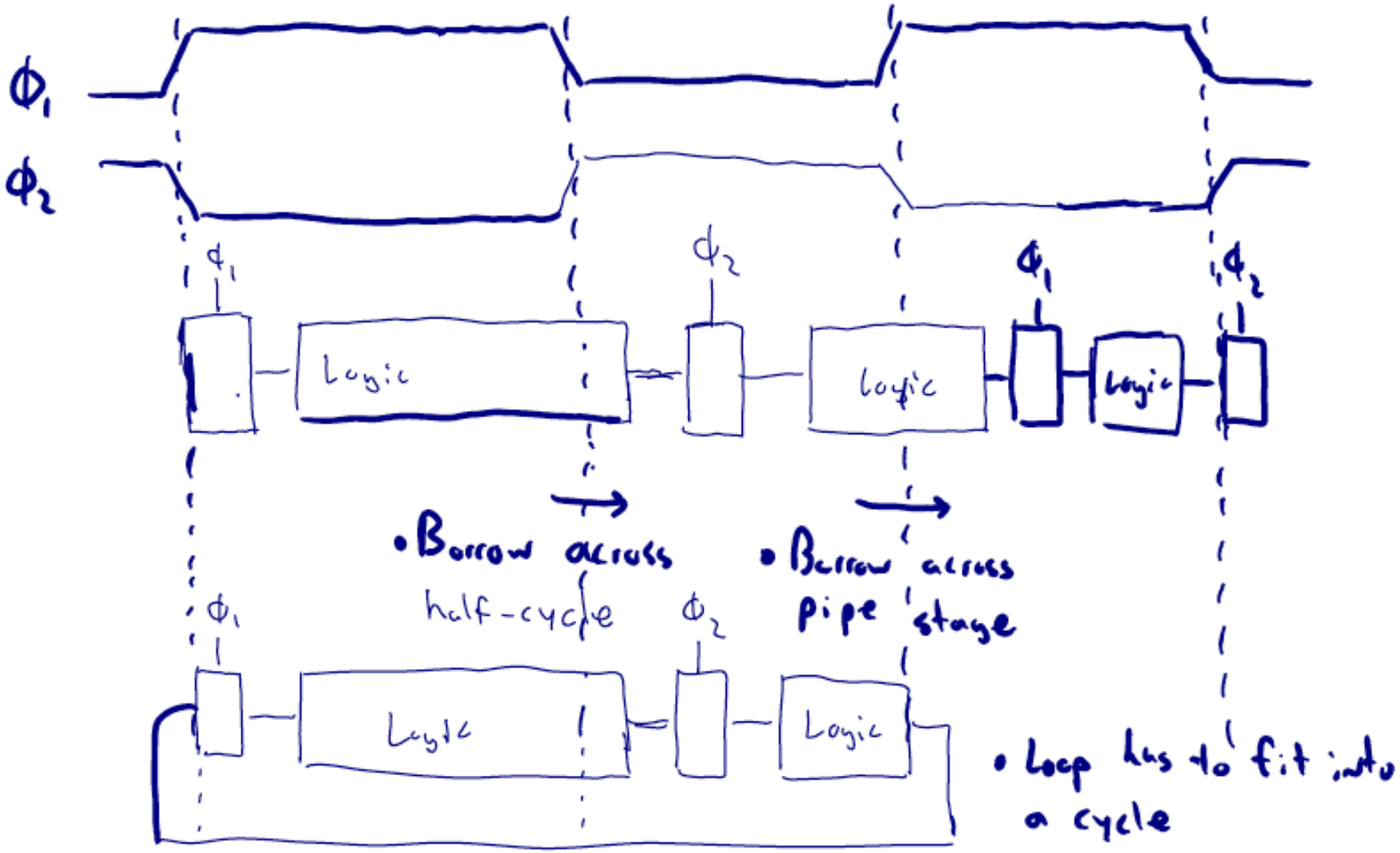




(Domino logic almost always uses latch-based clocking)

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Time-borrowing example



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Latch vs Flip-flop

- In a flip-flop based system
  - Data launches on one rising edge
    - And must arrive before next rising edge
  - If data arrives late, system fails
    - If it arrives early, wasting time
  - Flip-flops have hard edges
- In a latch-based system
  - Data can pass through a latch while it is transparent
  - Long cycle of logic can borrow time into next cycle
    - As long as each loop finished in one cycle

Latch vs Flip-flop

- Flip-flop generally easier to use
  - Most digital ASICs designed with register-based timing
- But, latches (both pulsed and level-sensitive) allow more flexibility
  - And hence can potentially achieve higher performance
  - Latches can also be made more tolerant of clock uncertainty